

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q15-0486

Eagle Rock 138 kV
June 17, 2016

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q15-0486. The Interconnection Customer proposes a 75.1 MW solar photovoltaic facility interconnecting at the Eagle Rock 138 kV substation in the Virginia area. The request was received on June 11, 2015. The study evaluated system intact and single contingency conditions using the regional planning case, including 280 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$16,273,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Travis Ab - Werner West 138.0 kV L
Contingency: Roanoke Rapids - Edgecombe Genco Llc 345.0 kV
Loading: 176.8% (Rating: 259 MVA)
DFAX: 9.9%
Pre-project loading: 173.9%
2. Facility: Fritch - Lakeside 138.0 kV
Contingency: Mitchell Dam - Skaggs Island Jct 2 345.0 kV line
Loading: 152.7% (Rating: 303 MVA)
DFAX: 5.1%
Pre-project loading: 151.5%
3. Facility: Haggin - Mitchell Dam 345.0 kV
Contingency: Hull St Road - Haggin 345.0 kV
Loading: 145.8% (Rating: 1,228 MVA)
DFAX: 17.2%
Pre-project loading: 144.8%
4. Facility: Spring St - Wing Lane 138.0 kV
Contingency: Skaggs Island Jct 2 - Mitchell Dam 345.0 kV
Loading: 138.1% (Rating: 309 MVA)
DFAX: 5.4%
Pre-project loading: 136.8%
5. Facility: Villa Del Rey 3010 - Floraday 138.0 kV L
Contingency: N Tower Tap - Mccartey 345.0 kV
Loading: 137.0% (Rating: 618 MVA)
DFAX: 5.3%
Pre-project loading: 136.4%

6. Facility: Bear - Fluvanna P.S 345.0 kV Line
Contingency: Leavitt - Birchwood 345.0 kV
Loading: 137.0% (Rating: 1,169 MVA)
DFAX: 20.4%
Pre-project loading: 135.7%
7. Facility: Seco - Wing Lane 138.0 kV Line
Contingency: Mitchell Dam - Skaggs Island Jct 2 345.0 kV L
Loading: 136.7% (Rating: 413 MVA)
DFAX: 6.8%
Pre-project loading: 135.5%
8. Facility: Mecklenburg Power Station - Kerr Dam 345.0 kV
Contingency: McCartney - N Tower Tap 345.0 kV
Loading: 132.5% (Rating: 1,198 MVA)
DFAX: 52.1%
Pre-project loading: 129.2%
9. Facility: Wells Ic - Geneva Border City 138.0 kV
Contingency: Ker Dam Substation - Mecklenburg Power Sta 345.0 kV Line
Loading: 130.7% (Rating: 315 MVA)
DFAX: 26.6%
Pre-project loading: 124.4%
10. Facility: Old Hickory Substation - Dewalt 138.0 kV
Contingency: McCartney - N Tower Tap 345.0 kV Line
Loading: 129.0% (Rating: 275 MVA)
DFAX: 9.6%
Pre-project loading: 126.4%
11. Facility: Natchez - Lakeside 138.0 kV line
Contingency: Mitchel Dam Substation - Skaggs Island Junction 2 345.0 kV line
Loading: 127.1% (Rating: 317 MVA)
DFAX: 5.1%
Pre-project loading: 125.9%
12. Facility: Westcoast Cellufibre - Leavitt 138.0 kV Line 1
Contingency: Mitchell Dam - Skaggs Island Junction 2 345.0 kV
Loading: 127.0% (Rating: 281 MVA)
DFAX: 5.7%
Pre-project loading: 125.5%
13. Facility: Westcoast Cellufibre - Leavitt 138.0 kV #2
Contingency: Mitchell Dam - Skaggs Island Junction 2 345.0 kV Line
Loading: 127.0% (Rating: 281 MVA)
DFAX: 5.7%
Pre-project loading: 125.5%
14. Facility: E Sparta - Gaston 138.0 kV
Contingency: Roanoke Rapids Substation - Edgecombe Genco Llc 345.0 kV
Loading: 126.7% (Rating: 204 MVA)
DFAX: 6.0%
Pre-project loading: 124.5%
15. Facility: Seco - Harvard 138.0 kV
Contingency: Mitchell Dam - Skaggs Island Junction 2 345.0 kV
Loading: 126.4% (Rating: 524 MVA)
DFAX: 6.8%
Pre-project loading: 125.4%
16. Facility: Hull St Rd - Harvard 138.0 kV L
Contingency: Mitchell Dam - Skaggs Island Junction 2 345.0 kV

- Loading: 123.1% (Rating: 574 MVA)
DFAX: 6.8%
Pre-project loading: 122.2%
17. Facility: Geneva Border City - Dewalt Substation 138.0 kV Line
Contingency: N Tower Tap - McCartney 345.0 kV
Loading: 122.3% (Rating: 302 MVA)
DFAX: 9.6%
Pre-project loading: 119.9%
18. Facility: Skaggs Island Junction 2 - Chancellor Row 345.0 kV line
Contingency: New Denver - Martin Dam 138.0 kV
Loading: 121.1% (Rating: 1,175 MVA)
DFAX: 14.9%
Pre-project loading: 120.2%
19. Facility: Fritch - Big Hanaford 138.0 kV line
Contingency: Mitchell Dam - Skaggs Island Jct 2 Substation 345.0 kV line
Loading: 120.2% (Rating: 529 MVA)
DFAX: 6.5%
Pre-project loading: 119.3%
20. Facility: N Tower Tap - Dpc Elroy 138.0 kV
Contingency: McCartney - N Tower Tap 345.0 kV
Loading: 119.1% (Rating: 561 MVA)
DFAX: 7.2%
Pre-project loading: 118.1%
21. Facility: Mccrtney - Roanoke Rapids 345.0 kV
Contingency: Gaton - McCartney Substation 345.0 kV Line
Loading: 115.2% (Rating: 1,398 MVA)
DFAX: 17.1%
Pre-project loading: 114.3%
22. Facility: Walthall - Unknown 138.0 kV
Contingency: North Tower Tap - Haggin 345.0 kV Line
Loading: 115.1% (Rating: 658 MVA)
DFAX: 7.2%
Pre-project loading: 114.3%
23. Facility: N Tower Tap - Unknown 138.0 kV
Contingency: Haggin - N Tower Tap 345.0 kV
Loading: 114.4% (Rating: 675 MVA)
DFAX: 7.2%
Pre-project loading: 113.6%
24. Facility: Hull Street Rd 345 kV/138 kV Transformer
Contingency: Mitchell Dam - Skaggs Island Jct 2 345.0 kV
Loading: 113.3% (Rating: 891 MVA)
DFAX: 7.7%
Pre-project loading: 112.6%
25. Facility: Roanoke Rapids - Gaston 345.0 kV
Contingency: Roanoke Rapids - McCartney 345.0 kV
Loading: 111.5% (Rating: 1,899 MVA)
DFAX: 7.9%
Pre-project loading: 111.2%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Travis Ab - Werner West 138.0 kV L	Rebuild line with higher-capacity conductor	\$7,883,000	16% (\$1,245,000)
Fritch - Lakeside 138.0 kV	Rebuild line with higher-capacity conductor	\$3,250,000	1% (\$32,000)
Mitchell Dam - Haggin 345.0 kV L	Rebuild line with higher-capacity conductor	\$26,238,000	1% (\$263,000)
Spring St - Wing Lane 138.0 kV	Rebuild line with higher-capacity conductor	\$1,441,000	1% (\$18,000)
Villa Del Rey 3010 - Floraday 138.0 kV	Rebuild line with higher-capacity conductor	\$7,872,000	1% (\$86,000)
Fluvanna P.S - Bear 345.0 kV L	Rebuild line with higher-capacity conductor	\$158,138,000	1% (\$1,487,000)
Seco - Wing Lane 138.0 kV Line	Rebuild line with higher-capacity conductor	\$1,548,000	1% (\$17,000)
Mecklenburg Power Station - Kerr Dam 345.0 kV	Rebuild line with higher-capacity conductor	\$41,882,000	2% (\$783,000)
Wells Ic - Geneva Border City 138.0 kV line	Rebuild line with higher-capacity conductor	\$10,338,000	7% (\$711,000)
Old Hickory - Dewalt 138.0 kV	Rebuild line with higher-capacity conductor	\$3,362,000	24% (\$806,000)
Natchez - Lakeside 138.0 kV line	Rebuild line with higher-capacity conductor	\$2,380,000	1% (\$23,000)
Westcoast Cellufibre - Leavitt 138.0 kV Line 1	Rebuild line with higher-capacity conductor	\$1,426,000	1% (\$15,000)
Leavitt - Westcoast Cellufibre 138.0 kV Circuit 2	Rebuild line with higher-capacity conductor	\$1,517,000	1% (\$16,000)
E Sparta Substation - Gaston 138.0 kV	Rebuild line with higher-capacity conductor	\$26,772,000	21% (\$5,619,000)
Seco - Harvard 138.0 kV	Rebuild line with higher-capacity conductor	\$4,230,000	1% (\$46,000)
Hull St Road - Harvard 138.0 kV line	Rebuild line with higher-capacity conductor	\$1,652,000	1% (\$18,000)
Geneva Border City - Dewalt Substation 138.0 kV Line	Rebuild line with higher-capacity conductor	\$15,936,000	24% (\$3,822,000)
Chancellor Row - Skaggs Island Jct 2 345.0 kV Line	Rebuild line with higher-capacity conductor	\$26,239,000	1% (\$174,000)
Fritch - Big Hanaford 138.0 kV line	Rebuild line with higher-capacity conductor	\$4,677,000	1% (\$40,000)
Dpc Elroy - North Tower Tap 138.0 kV	Rebuild line with higher-capacity conductor	\$4,514,000	2% (\$95,000)
Mccrtney - Roanoke Rapids 345.0 kV	Rebuild line with higher-capacity conductor	\$43,552,000	1% (\$409,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Walthall Substation - Unknown 138.0 kV	Rebuild line with higher-capacity conductor	\$20,325,000	1% (\$243,000)
N Tower Tap - Unknown 138.0 kV	Rebuild line with higher-capacity conductor	\$5,183,000	1% (\$62,000)
Hull Street Rd 345 kV/138 kV Transformer	Replace transformer with higher-rated unit	\$15,955,000	1% (\$128,000)
Gaston - Roanoke Rapids 345.0 kV	Rebuild line with higher-capacity conductor	\$28,387,000	0% (\$115,000)
	Total allocated to project		\$16,273,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.